

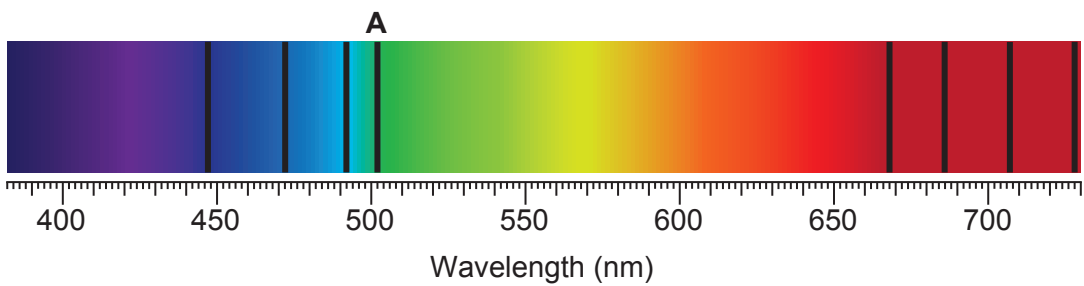
6. The following equation shows a fusion reaction that happens in stars.



(a) State the **two** conditions needed for a fusion reaction to take place in a star. [2]

- 1.
- 2.

(b) An incomplete absorption spectrum from the Sun is shown below.



(i) An absorption line at wavelength $590 \times 10^{-9} \text{ m}$ is missing from the spectrum. **Complete the diagram** above by adding the missing absorption line. [1]

(ii) Describe **and** explain how the position of the absorption line labelled **A** would be different for a star in a more distant galaxy. [3]

.....

.....

.....

.....

.....

